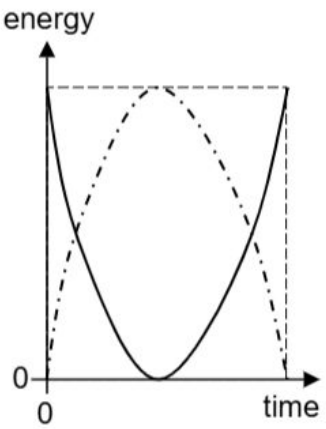
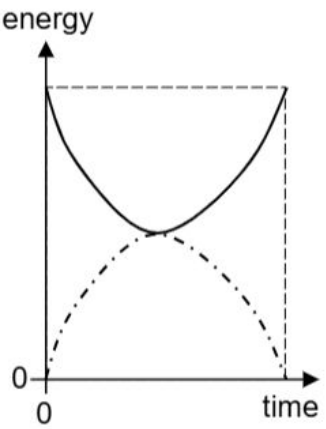
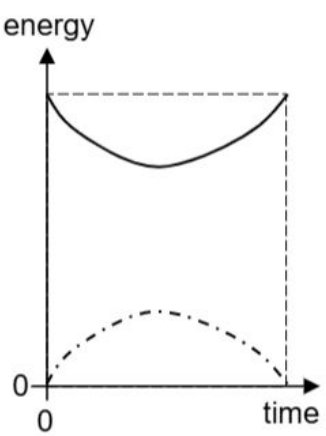
8	A golfer hits a golf ball so that it leaves the club at an angle of 60° to the horizontal. Assume that air resistance can be ignored. Which graph best represents the variation of kinetic energy and of gravitational potential energy of the ball during the time of flight?		
	A	**B**	



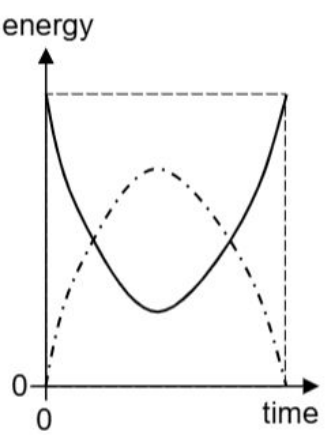


key
kinetic energy
gravitational potential energy

C



D



L3 **Answer: D**

In 2-D projectile motion, horizontal velocity is non-zero and constant throughout.
➔ **Eliminate option A.**

Let u be the initial speed.
Initial KE = ½ m u^2 = Total energy, E

Horizontal velocity component = u cos θ, where θ is the angle above the horizontal.

When θ = 60°, Horizontal velocity component = u cos 60° = ½ u
KE at maximum height where vertical velocity equals zero and horizontal velocity remains
= ½ m (½ u)^2 = ¼ (½ m u^2) = **¼ E**
By conservation of total energy, **GPE at maximum height** = E – ¼ E = **¾ E**
➔ **Answer is option D.**

Additional notes:

When θ = 45°, Horizontal velocity component = u cos 45° = $\frac{1}{\sqrt{2}}$ u

KE at maximum height = ½ (½ m u^2) = **½ E**
By conservation of total energy, **GPE at maximum height** = **½ E**
➔ **Option B.**

In summary,

When θ = 45°, object will retain EXACTLY HALF of its initial KE at maximum height, hence KE and GPE graphs will TOUCH AT THE TURNING POINTS e.g. Option B.

When θ < 45°, object will retain MORE THAN HALF of its initial KE at maximum height, hence KE and GPE graphs will NOT overlap e.g. Option C.

When θ > 45°, object will retain LESS THAN HALF of its initial KE at maximum height, hence KE and GPE graphs WILL overlap e.g. Option D.

When θ = 90°, i.e. vertical motion in 1-D, object will come to instantaneous rest (KE = 0) at maximum height, e.g. Option A.

Why are KE and GPE graphs parabolic shaped?

GPE = mgs_y where $s_y = u_y t + \frac{1}{2}(-g)t^2$ → GPE-t graph is a negative quadratic function.

KE = TE – GPE = constant – (-ve quadratic function) = +ve quadratic function.